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4. Write short notes on the following.
   a) Unit
   b) Rounding - off
   c) Dimensionless quantities

5. Explain the principle of homogeniety of dimensions. What are its uses? Give example.

IV. Numerical Problems

1. In a submarine equipped with sonar, the time delay between the generation of a pulse and its echo after reflection from an enemy submarine is observed to be 80 s. If the speed of sound in water is 1460 ms^{-1}. What is the distance of enemy submarine? Ans: (58.40 km)
2. The radius of the circle is 3.12 m. Calculate the area of the circle with regard to significant figures. Ans: (30.6 m^2)
3. Assuming that the frequency γ of a vibrating string may depend upon i) applied force (F) ii) length (ℓ) iii) mass per unit length (m), prove that $\gamma \alpha \frac{1}{l}\sqrt{\frac{F}{m}}$ using dimensional analysis. (related to JIPMER 2001)
4. Jupiter is at a distance of 824.7 million km from the Earth. Its angular diameter is measured to be 35.72″. Calculate the diameter of Jupiter. Ans: (1.428×10^5 km)
5. The measurement value of length of a simple pendulum is 20 cm known with 2 mm accuracy. The time for 50 oscillations was measured to be 40 s within 1 s resolution. Calculate the percentage accuracy in the determination of acceleration due to gravity 'g' from the above measurement. Ans: (6%)

V. Conceptual Questions

1. Why is it convenient to express the distance of stars in terms of light year (or) parsec rather than in km?
2. Show that a screw gauge of pitch 1 mm and 100 divisions is more precise than a vernier caliper with 20 divisions on the sliding scale.
3. If humans were to settle on other planets which of the fundamental quantities will be in trouble? Why?
4. Having all units in atomic standards is more useful. Explain.
5. Why dimensional methods are applicable only up to three quantities?

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